

claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

```
- Source: `claude-wsl-ubuntu`
- Kind: `claude`
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d\subagents\workflows\wf_fa7a2077-3a9\agent-a8beca64e2ef9d8aa.jsonl`
- SHA-256: `2fc3dfca27dd50ad3ed2067bcfb237cde06c805ee4b49a33cf3cbb152e204f66`
- Source modified: `2026-06-29T10:51:16+00:00`
- Imported at: `2026-07-05T16:48:30+00:00`
- project: `wf_fa7a2077-3a9`
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`
```

Transcript

1. user (2026-06-29T10:50:03.906Z)

Read /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/HOW_RALLY_DETECTION_WORKS.md and /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer/docs/CODE_MAP.md if they exist. Summarize the rally-detection pipeline mental model and the navigation map, and explain what the "served" operating point / preset means in the A/B context. If either file is missing, say so. Return data, not prose.

2. assistant (2026-06-29T10:50:06.552Z)

I'll read both files to summarize the rally-detection pipeline and navigation map.

3. user (2026-06-29T10:50:07.175Z)

```
1      # How rally detection works (the 2-layer mental model)
2
3      A short conceptual map of how we go from raw video to rally `[start, end]` segments ?
4      and, crucially, **which layer our annotation can improve.**
5
6      ...
7      video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
8              ?
9              (WASB CNN, per-frame trajectory
10             frozen weights) ?
11             ?
12             velocity signal (how fast the shuttle moves)
13             ?
14             [SEGMENTATION] ??? the 3 knobs we tune
15             ?
16             ?
17             rally windows [start, end] ??? IoU vs golden
labels
18      ...
19
20      ## Two layers ? keep them separate in your head
21
22      **1. DETECTION ? "where is the shuttle, this frame?"**
23      WASB is a CNN that, for each frame, outputs a heatmap ? the shuttle's `(x, y)` + a
24      visibility/confidence. Its weights were learned (by gradient descent, on shuttle-
coordinate
25      labels) and we use them **frozen**. We chose a *shuttle* detector on purpose: "shuttle
in
26      flight" is the signal for "a rally is happening," and it's **camera-agnostic** ? unlike
27      player-motion, which fooled the old heuristic on broadcast cameras that pan with the
players.
28      (Your videos are static tripod, so player-motion would also work ? but shuttle-motion is
```

```

the
29     robust general bet.)
30
31     **2. SEGMENTATION ? "when is a rally?"**
32     From the per-frame `(x, y)` we compute the shuttle's **velocity** (displacement between
frames).
33     Then three knobs turn that into rally intervals:
34     - **`velocity_thresh`** ? a frame is "active" only if the shuttle is moving faster than
this
35         (in flight, not lying on the floor / being picked up).
36     - **`merge_gap`** ? active frames within this many seconds get joined into one rally
(bridges
37         brief gaps: a shuttle momentarily occluded, or the beat between two hits).
38     - **`min_window_duration`** ? throw away windows shorter than this (blips / false
flickers).
39
40     Output: a list of `[start, end]` windows = the rallies.
41
42     ## Where the annotation fits (the key point)
43     Our Rooru/golden labels are **rally `[start, end]` times** ? they say *when* rallies
are, not
44     *where* the shuttle is*. So they can't teach WASB's detector (that needs `(x,y)` labels).
They
45     **can** grade and tune the **segmentation layer**: we score predicted windows against
the golden
46     rallies with **temporal IoU** (how much they overlap), and pick the 3 knobs that
maximise it ?
47     **calibrate on some rallies, measure on held-out ones** so the number is honest.
48
49     ## Where sparks could land (open questions ? not commitments)
50     Things worth a shower-thought; any of these could beat 3 hand-tuned thresholds:
51     - **Richer features than raw velocity:** acceleration, trajectory *continuity* (a real
rally is a
52         smooth arc that survives occlusion), shuttle *height/arc*, or **direction reversals**
(each
53         reversal ? a shot ? ties to `shots_count`!).
54     - **A tiny learned segmenter** instead of 3 thresholds: feed the trajectory features to
a small
55         model trained on the golden rally labels (still the segmentation layer, just smarter).
56     - **Hysteresis** (two thresholds: enter-rally high, stay-in-rally low) ? robust to mid-
rally dips.
57     - **Rhythm/periodicity:** rallies have a hit cadence; dead time doesn't.
58     - **Fusion:** combine shuttle-motion (WASB) with player-motion for static cams. ?
designed in
59         [MULTI_SIGNAL_FUSION_PLAN.md](MULTI_SIGNAL_FUSION_PLAN.md) (one rally-fusion layer;
the person cue is
60         ADR-007's parked #2 and `rally_gate.py`'s "Gate B" ? now built (default-OFF,
unmeasured on golden) in `fusion_hybrid.py`).
61     - **Per-video calibration as a *feature*:** "watch a few labeled rallies ? auto-tune to
this
62         court/lighting ? segment the rest sharply" ? exactly your earlier idea, at the right
layer.
63
64     The whole point of per-video calibration: a handful of *your* labeled rallies nudges
these knobs
65     to *your* court/camera, and we report the IoU lift on the rallies it never saw.
66

```

4. user (2026-06-29T10:50:08.410Z)

<system-reminder>[Truncated: PARTIAL view ? showing lines 1-205 of 386 total (39849 tokens, cap 25000). Call Read with offset=206 limit=205 for the next page, or Grep to find a specific section. Do NOT answer from this page alone if the answer may be further in the file.]</system-reminder>

```

1      # CODE_MAP.md
2
3      **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-
hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a
segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.
4
5      **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this
doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure
(this revision follows #40).
6
7      ? **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust
git history over stale lines until the next full regen:** `add_video` is now an **UPSERT** (no
FK-cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via
`segment_watermarks`/`prune_segments`); detector artifacts are keyed by **video_id**, not
basename; **indexing.wasb.** is now a real typed config** (was silently dropped); `timeout_sec`
defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**;
`match_intervals` requires real overlap (iou>0); `run_regression` baseline lives in tracked
`eval_baselines/` and **exits 3 on a missing baseline**; the API validates
`video_path`/`output_filename`. New tests: `test_api_endpoints`, `test_compiler`,
`test_wasb_runner`, `test_reingest_safety`, `test_detector_keying`, `test_config_wasb`,
`test_path_safety`, `test_reliability_hardening` (+ collector `test_licensing_gate`,
`test_import_guard`, `test_cvat_robustness`).
8
9      **LEGEND**
10     - `@path` = file (always repo-relative)
11     - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
12     - `->` = dataflow / call / "produces"
13     - `$` = data contract (canonical shape)
14     - `?` = gotcha / footgun
15     - `?` = registry / plugin extension point
16     - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
17
18     ---
19
20     ## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
21
22     Top-level tree and the **placement rule** for each kind of file. When unsure, prefer an
existing
23     **registry / extension point** (`?`) over a new top-level file.
24
25     | You're adding? | Put it in? | Notes |
26     |---|---|---|
27     | Backend / runtime code | `backend/<subsystem>/` |
`pipeline/{segmenters,detectors,validators}`, `providers/`, `sports/`, `annotations/`,
`infrastructure/`, `reporting/`, `stitcher/`, `utils/`, `config/`, `features/`, `api/`. Most
additions are **register a class in the relevant `?` registry**, not a new module tree. |
28     | **Eval LIBRARY** (importable, pure ? metrics, calibration, a scorer, a feature) |
`backend/eval/` | Pure core, no side-effects at import; CLI/I-O lives in a separate driver.
These are **imported widely** (e.g. `calibrate_wasb` by 8 sites) ? they are library, not
scripts; do NOT move them out. |
29     | **Standalone run-driver / one-off entry CLI** (has `__main__`, NOT imported by app
code) | **scripts/** | e.g. `scripts/run_phase3_eval.py`, `scripts/run_process_and_stitch.py`.
Run as `python scripts/<name>.py`. If it needs `load_dotenv()`/`sys.path` before importing
backend, add `["scripts/<name>.py"] = ["E402"]` to `ruff.toml`. |
30     | **Ops / maintenance tool** (not part of the served app ? cache GC, batch admin) |
`tools/` | Run as `python -m tools.<name>` (it's an importable package). e.g.
`tools/cleanup_caches.py`. Keep the destructive decision path pure + unit-tested. |
31     | Training code (model fitting) | `training/` | Behind the CI-enforced import wall
(ADR-008) ? `backend.main` must not import it. Own `requirements-train.txt`. |
32     | A test | `tests/test_*.py` | pytest auto-discovers; the full-suite lane has a **70%
coverage floor**. Pure/offline + mocked (no GPU/WSL/network). |
33     | A doc (design, plan, living log) | `docs/` | Index it in `docs/README.md`. Only
**terminal-state** (DONE/REJECTED) work moves to `docs/archives/`. |
34     | A throwaway repro / debug script | `scratch/` | **Gitignored** + excluded from
ruff/pytest. Never imported by app code. |

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35 | A model artifact (weights + manifest) | `artifacts/<name>/<ver>/` | ADR-008:
**manifest committed** (provenance), **weights gitignored**. |
36 | Eval baseline / leaderboard JSON | `eval_baselines/` | Tracked (survives a clean
checkout); the no-regression gate reads it. |
37 | **Committed regression fixture** (video-free golden) |
`eval_baselines/fixtures/<slug>/` | Tracked, **LF-pinned** (`.gitattributes`). Synthetic Tier-0
(or owner-cleared, de-attributed real). Drives the `golden_fixtures` characterization gate with
**no video/GPU/DB**. See `docs/GOLDEN_REGRESSION_FIXTURES.md`; mint via `python -m
backend.eval.golden_fixtures --freeze-synthetic` / `--freeze-real` (refusal-gated). |
38 | Pipeline run outputs (reels, trajectories, DBs, frames) | `output/` | **Gitignored.**
Per-video isolation is the tracked **`PER_VIDEO_WORKSPACE.md`** project (P0 helper shipped #264;
supersedes the archived `PER_VIDEO_OUTPUT_LAYOUT`). |
39
40 **Canonical files that stay at the repo ROOT** (referenced by CI / packaging / docs ? do
not move):
41 `pyproject.toml` (PEP 621 packaging, hatchling; `indexer` entry point) .
`run_all_tests.py` (CI: `ci.yml`) .
42 `run_regression.py` (the no-regression gate CLI, cited in `EXPERIMENT_HARNESS.md`) .
`run_eval.py`
43 (grandfathered ? imported by `tests/test_eval_harness.py`). The old diagnostic
`setup.py` moved to
44 `scripts/doctor.py` (it shadowed the build system); invoke via `python
scripts/doctor.py` or `indexer --setup`.
45 New standalone run-drivers go in **`scripts/`**, not the root.
46
47 ---
48
49 ## 1. PIPELINE
50
51 ### 1A. Runtime: video file -> highlight reel
52 ```
53 typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-
defaults, never fatal)
54 @backend/config/loader.py::load_config ->
@backend/config/models.py::AppConfig
55 -> video_id = MD5(whole file) @backend/utils/hashing.py::compute_video_id
(single shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint
copies)
56 -> validate
@backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)
57 FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity
-> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see
@backend/pipeline/validators/_init_.py)
58 [validation FAIL -> persist status="failed", return early; report STILL emitted
in finally]
59 -> db.add_video(video_id, filepath, status, [r.dict()])
@backend/infrastructure/database.py
60 -> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback
'motion')
61 -> seg = segmenter_registry.get(seg_name)(db, global_config) ?
@backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)
62 -> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
63 [STOP POINT: segments-only ? predictions now in DB; eval/regression operate here
without stitching]
64 -> finally: emit_indexer_report(...) @backend/reporting/_init_.py (? ALWAYS runs
? even on validation failure / exception; NEVER raises; grafts response["reports"])
65 -> select segment ids -> [(start,end)] (+ stitching padding)
66 -> VideoCompiler.compile_highlights(filepath, time_pairs, out_name)
@backend/pipeline/stitcher/compiler.py
67 per-clip ffmpeg re-encode (libx264/superfast/crf22) -> concat -c copy ->
output/<name>.mp4
68 ```
69 Hybrid segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same
shape):
70 ```

```

```

71     P1 chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)
72     -> P2 candidate "action windows" from motion/trajectory [STOP: candidates persisted
via db.add_candidate_segment]
73     -> P3 pad clip (ai_handoff_padding) -> ThreadPoolExecutor ->
provider.analyze_video(clip, prompt) ? provider_registry
74     -> P4 db.add_play_segment + db.link_candidate_to_segment
75     ```
76     Pure-AI path `gemini`: no P2; chunk-or-single -> provider -> heavy validate
(clamp/dedup) -> add_play_segment.
77     Legacy `motion`: pixel-diff -> segments + **fabricated** metadata/events (? not ground
truth).
78
79     ### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
80     ```
81     WasbHybridSegmenter.process_video
@backend/pipeline/segmenters/wasb_hybrid.py
82     -> WasbRunner.healthcheck() (gate; not ok -> return [])
@backend/pipeline/detectors/wasb_runner.py
83     -> WasbRunner.run_predict(video, out_dir)
84     stage video + wasb_infer.py into WSL fs -> `wsl bash -c 'source conda.sh && python
wasb_infer.py ...'`
85     -> @backend/pipeline/detectors/wasb_infer.py::run (WSL): sliding windows -> GPU
detector pass (CHECKPOINTED det_raw.jsonl) -> CPU tracker (always full re-run) -> trajectory CSV
86     -> TrackNetRunner.parse_trajectory_csv(csv)
@backend/pipeline/detectors/tracknet_runner.py (shared, re-exported by WasbRunner)
87     -> TrackNetRunner.trajectory_to_action_windows(points, fps, frame_width, chunk_start,
velocity_thresh, merge_gap, min_window_duration, chunk_index) [the model-agnostic WINDOWING
BRAIN]
88     ```
89     (`tracknet_hybrid` identical, swapping WasbRunner->TrackNetRunner; `_probe_fps_width`
fallback (30.0, 1280.0).)
90
91     ### 1C. Calibration / eval flow (NO pipeline run unless told)
92     ```
93     GT CSV -> annotations.load_annotations / calibrate_wasb.load_golden_gts ->
List[Interval]
94     DB predictions -> harness.get_predictions(db, video_id, stage) -> List[Interval] stage
? {candidates, final}
95     -> metrics.match_intervals (greedy temporal-IoU) -> metrics.score -> Score
96     -> harness.evaluate -> EvalReport -> report_to_dict
97     -> batch.aggregate (corpus micro/macro) @backend/eval/batch.py
98     -> regression.regress(_with_provider) vs baseline @backend/eval/regression.py
[CI gate]
99     Calibration: calibrate_wasb.run -> calibration.{select_best, improvement_report,
cross_validate} @backend/eval/calibration.py
100    Distill Gemini->local: calibrate_local (thresholds) OR distill_local (learned
classifier) @backend/eval/{calibrate_local,distill_local}.py
101    Learned segmenter (offline): training.build_training_set(X,y) ->
classifier.LogisticRegression.fit/save
102    Gemini refinement driver (tuning, standalone):
gemini_refine.{refine_window,full_video_rallies} @backend/eval/gemini_refine.py
103    Audio probe (diagnostic only, NOT in live pipeline): audio/e1_probe.run
@backend/eval/audio/e1_probe.py
104    ```
105    Entrypoints: `@run_eval.py` (single video score), `@scripts/run_phase3_eval.py`
(manifest batch, can segment), `@run_regression.py` (baseline gate, exit 1=regression),
`@scripts/nightly_regression.py` (nightly CACHED-trajectory CPU rally-quality tripwire, exit
1=regression/4=stale), `@scripts/nightly_reelcount.py` (nightly FULL-pipeline reel-count +
quality + cost on a GCP GPU VM, emailed), `@backend/eval/calibrate_wasb.py` +
`@backend/eval/export_wasb_segments.py` (WASB tuning drivers).
106
107    ---
108
109    ## 2. SUBSYSTEMS
110

```

```

111     ### 2.0 Orchestration & config
112     - `@backend/main.py` ? Thin CLI + FastAPI bootstrap (orchestrator; modes via FLAGS).
Static UI mounts `/static` from `backend/api/static`. Logging is initialized with `basicConfig`
(INFO, timestamped format); backend modules use `logger = logging.getLogger(__name__)` (replaces
stray prints in hot paths like motion, stitcher, wasb_infer, config loader). User-facing CLI
summaries may still use `print` for direct output.
113     - `::app` FastAPI; `::db_instance`, `::global_config` (`Optional[AppConfig]`) module
globals set ONLY in `main()`. ? importing `app` for ASGI without `main()` -> both `None` ->
every endpoint NPEs.
114     - **ASYNC (#16, PR-1):** `::process_video` `@POST /api/process` -> fast-fail 4xx
(path/exists/segmenter) -> `db.create_job` -> `_get_executor().submit(_drain_due_jobs)` -> **202
+ job_id**. `::drain_due_jobs` / `::run_claimed_job` (worker loop EXTRACTED to
`@backend/api/job_worker.py` (#234), re-exported on `backend.main`): claim each runnable job via
`db.claim_due_job` -> `_execute_processing` -> mark done/failed (a `JobWaiting` parks the job
via `set_job_waiting` for resume; EXECUTION_POLICY P3 self-scheduling). `::execute_processing`
= the former sync body (hash -> validate -> add_video -> segment -> `finally:` always
`emit_indexer_report`, grafts `response["reports"]`; never raises from the report); a RAISING
segmenter now prune_after-restores pre-run rows before re-raising. `::get_job` `@GET
/api/jobs/{job_id}` -> `{status, progress, video_id, error, result, reports, ...}` ? job
`status` = EXECUTION state (`queued|running|done|failed`); the pipeline verdict lives in
`result["status"]`. `::get_executor` lazy module-global `ThreadPoolExecutor(max_workers=1)`
(single box + Gemini serial-upload learning; tests monkeypatch it inline). `main()` runs
`db.fail_stale_jobs()` at startup (orphaned queued/running -> failed). `::compile_highlights`
`@POST /api/compile`. `::query_play_segments` `@POST /api/query`. `::get_config` `@GET
/api/config`. `::run_cli_diagnose_clip` `--debug-clip`: lazy `import cv2`, samples  $\pm 3s$  vs
`indexing.motion_threshold` (dual model/dict read). ? CLI `--video-path` mode still runs the
pipeline synchronously inline (separate code path, unchanged by #16).
115     - **UPLOAD/DOWNLOAD (PR-2, B2+B3):** the chunked-upload endpoints now live in
`@backend/api/uploads.py` (#228 ? an `APIRouter` registered via `app.include_router`;
`uploads.py::upload_cfg` resolves the live `backend.main.global_config` lazily so it reads the
startup config, and a lazy `import backend.main` avoids an import cycle) ?
`uploads.py::upload_init` `@POST /api/upload/init` (`{filename, total_size_bytes?}` -> ext
allowlist + size gate -> `{upload_id}`), `::upload_part` `@PUT /api/upload/{id}/part/{n}` (raw-
bytes body, STRICTLY sequential n, streams to `.partial-<id>` under `security.upload_dir`; per-
part/total cap breach -> 413 + upload ABORTED), `::upload_complete` `@POST .../complete`
(`{total_parts}` -> assemble -> never-overwrite rename -> `{path, video_id(MD5), size_bytes}` ?
client then POSTs /api/process with that path), `::upload_abort` `@DELETE /api/upload/{id}`. ?
upload state (`uploads.py::uploads` + `::uploads_lock`) is IN-PROCESS ? restart aborts
partials (client re-inits); per-user partitioning lands with PR-3. `main.py::download_highlight`
`@GET /api/highlights/{video_id}/{filename}` (download was NOT moved ? still in main.py) ? safe-
name + `resolve_under_root(output_dir)` then FileResponse stream (the old /api/compile alert
path was browser-unreachable). Config: `SecurityConfig.{upload_dir='output/uploads',
max_upload_mb=4096, max_part_mb=95, allowed_upload_extensions=[mp4,mov]}` (95MB: free-tier edge
proxies cap bodies ~100MB).
116     - `::load_config` ? ? legacy thin wrapper returning a **dict**
(`load_app_config(path).model_dump(...)`); kept for transition ? new code imports
`load_app_config` `AppConfig` directly. `::ProcessRequest`/`::QueryRequest`/`::CompileRequest`
pydantic bodies.
117     - `main()` sets `global_config.output.output_dir = args.output_dir` (typed field on
`OutputConfig`); `db_path = os.path.join(global_config.output.output_dir, output.db_name)`.
`compile_highlights` reads the SAME `global_config.output.output_dir`, so `--output-dir` is
honored end-to-end (the old write-only `_runtime_overrides` hack is gone).
118     - ? `--segmenter` argparse `choices` frozen at build time from
`segmenter_registry.list_available()` ? a lazily-registered segmenter wouldn't appear. CLI
default_segmenter fallback `motion`.
119     - `@scripts/run_process_and_stitch.py` ? one-shot demo; ? hardcoded paths, interactive
`input()` (blocks automation), skips validation. Config via `backend.config.load_config`
(typed); segmenter, padding, and db path all read from the model (no literal fallbacks).
120     - `@run_eval.py::main` ? score one video's DB preds vs GT CSV.
`--stage{candidates,final,both}`. Exit 2=arg/IO. ? pure scorer; errors if video never processed.
`::default_db_path` resolves via `backend.config.load_config`
(`output.output_dir`/`output.db_name`). `get_file_md5` is an alias for `compute_video_id`.
121     - `@scripts/run_phase3_eval.py::main` ? walk manifest, segment-if-needed, score,
aggregate. `load_dotenv()` at import. ? DB key = file MD5, NOT manifest's `video_id` (kept as
label); config via `backend.config.load_config`; `segmenter_cls(db, cfg)` receives the typed

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```

`AppConfig` (same object the live pipeline feeds segmenters).
122 - `@run_regression.py::main` ? `regress_with_provider` vs baseline. Exit codes semantic:
**0=ok, 1=REGRESSION (CI gate), 2=missing DB**. ? `--update-baseline` runs AFTER compare
(snapshots even a regressing run); `::default_db_path` resolves via
`backend.config.load_config`.
123 - `@scripts/nightly_regression.py::main` ? **nightly rally-quality tripwire** (the dual-
eval-set discipline made automatic). Re-scores the golden corpus
(`output/human_lovo_manifest.json`) under TWO configs ? `default` (SERVED, milestone knobs OFF)
+ `milestone` (R3 cap=6 + R4 inactivity trim) ? via `backend.eval.rally_seg_eval.evaluate` (CPU,
seconds; CACHED trajectories, **no GPU/Gemini/WASB**), diffs headline metrics (`tol_f1`, tIoU
`f1@0.5/0.25`, `merge_rate`) + the **per-video over-seg guard** (`split_count`/`merge_count` may
not rise on ANY video) vs frozen `eval_baselines/nightly_baseline.json`. Pure core
(`summarize_run`/`compare_config`/`compare_all`/`format_report`) is unit-tested; thin I/O shell
scores + writes `output/nightly_regression/<date>.{md,json}`. Exit **0=pass, 1=regression,
2=missing corpus, 3=no baseline, 4=stale (corpus/config changed ? re-snapshot)**. `--update-
baseline` snapshots HEAD. **Default-ADDITIVE ? does NOT change the promotion gate.** Wired as a
local Windows Scheduled Task via `scripts/nightly_regression.ps1` (wrapper) +
`scripts/register_nightly_task.ps1` (a cloud agent can't ? the cached trajectories live outside
the repo). The future **RAW eval set** (currently-bad videos) plugs in as a third config. See
`docs/R2_EVAL_METRIC_DESIGN.md` + `memory/eval-dual-set-regression`.
124 - `@scripts/nightly_reelcount.py::main` ? **nightly FULL-pipeline engine regression**
(the GCP-VM sibling of nightly_regression; runs the REAL configured+checkpointed WASB pipeline,
not a cached re-score). Per manifest video (`eval_baselines/reelcount_manifest.json`, gs://
clips): `segmenter_registry.get(seg)(db,cfg).process_video` ? **reel count** =
`len(play_segments)` + (if labels) R2 quality (`rally_seg_eval.score_intervals`); diffs vs
`eval_baselines/reelcount_baseline.json` ? reel-count **EXACT** (`random.seed(0)` + re-run-once
guard for the transient `add_play_segment`?None flake), quality with tolerance; **cost** = VM-
uptime × $/hr (`backend.billing`, USD+INR). Pure core
(`summarize_results`/`compare`/`cost_report`/`format_report`) is unit-tested
(`tests/test_nightly_reelcount.py`); `scripts/nightly_email.py` emails the report (SMTP creds
from Secret Manager/env, graceful no-creds skip). Exit **0/1/2/3/4** (mirrors
nightly_regression). Runs on an ephemeral GCP GPU VM that self-deletes ?
`deploy/gcp/nightly_regression/` (`run_on_vm.sh` startup-script, `launch.sh` driver-agnostic,
`register_scheduler.sh` Cloud Scheduler?Cloud Run job). GPU/email paths owner-verified (not in
CI). See `deploy/gcp/nightly_regression/README.md` + `docs/NIGHTLY_ENGINE_REGRESSION.md`.
125
126 - `@backend/config/__init__.py` ? ? canonical config import surface. Re-exports
`load_config`, `save_default_config` (from `.loader`) + `AppConfig`, `Config`, `IndexingConfig`,
`StitchingConfig`, `ValidationConfig` (from `.models`); `__all__` = exactly those 7. Use `from
backend.config import load_config, AppConfig`.
127 - `@backend/config/models.py` ? Pydantic v2 models; **single source of truth for
defaults**.
128 - `::AppConfig` ? root. Scalars `sport='badminton'`, `ai_provider='gemini'`,
`ai_model='gemini-2.5-flash'`; sections `validation`, `indexing`, `output`, `stitching`
(`default_factory`). `model_config` = {extra:'allow', validate_assignment:True}. ?
`extra='allow'` -> unknown/typo'd config.json keys silently kept (legacy tolerance).
129 - ? `::AppConfig.get` ? legacy dict-compat shim: returns nested **sections as plain
dicts** (`model_dump(mode='json')`, NOT the model) so chained
`config.get("indexing",{}).get(...)` keeps working; falls back to searching a leaf key in any
section (sections discovered dynamically from `model_fields`). ? "bit every validator on real
runs" per docstring ? returning the model breaks `.get`. `::AppConfig.__getitem__` raises
`KeyError` for unknown top-level keys else delegates to `.get`.
130 - `::IndexingConfig` ? segmenter/detector knobs (`motion_threshold=0.08`,
`min_segment_duration=3.0`, `dead_time_merge_gap=2.0`, `chunk_duration=90.0`,
`chunk_overlap=10.0`, `yolo_velocity_threshold=0.15`, `yolo_frame_skip=3`,
`ai_handoff_padding=2.0`, `ai_max_workers=5`, `log/store_yolo_candidates=True`, nested
`tracknet`). ? `default_segmenter='yolo_hybrid'` HERE but config.json overrides to `gemini`. ?
`::IndexingConfig.segmenter_must_be_registered` `@field_validator` is a **no-op pass-through** ?
a typo'd segmenter validates fine, only fails later at `segmenter_registry.get(...)`).
131 - `::ValidationConfig` (`min_resolution_width=1280`, `min_resolution_height=720`, `min_f
ps=29.9`, `min_blur_threshold=20.0`, `max_scene_cuts_per_minute=0.5`, nested `relevance`);
`::ValidationRelevanceConfig` (court HSV gates, `court_pixels_min_ratio=0.05`).
`::TrackNetConfig` (WSL invocation contract: `wsl_distro='Ubuntu'`, `repo_dir`,
`conda_env='TrackNetV4'`, `weights_path='', `queue_length=5`, `velocity_threshold=0.02`).
`::OutputConfig` (`default_highlights_name`, `db_name='sports_indexer.db'`,

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`output_dir='output'` ? CLI `--output-dir` overrides it on the typed model in `main()`).
`::StitchingConfig`
(`begin_padding=0.5`,`end_padding=0.5`,`use_cache=False`,`min_shot_count=1`). `::Sport` Literal
of 5 sports. `::Config = AppConfig` alias.
132     - `@backend/config/loader.py` ? load/save with defaults-merge.
133     - config path = `backend.utils.app_home.default_config_path()` (P0a; honours
`RALLY_APP_HOME`, else `Path("config.json")` CWD-relative ? the old `DEFAULT_CONFIG_PATH`
constant was dropped as a frozen-at-import footgun). `::_get_builtin_defaults` =
`AppConfig().model_dump(...)` .
134     - `::load_config` ? precedence built-in-defaults -> file overrides. ? **shallow top-
level merge** (`{**defaults, **file_data}`): a section in config.json REPLACES the whole
defaults dict for that section (Pydantic then re-applies field defaults for missing leaves).
Missing/unreadable/parse-error file -> fully-defaulted AppConfig + `[CONFIG WARNING]` (non-
fatal).
135     - `::save_default_config` ? writes fresh `config.json`, **overwrites** if present
(used by scripts/doctor.py/--setup). `::get_default_config` ? ? transitional plain-dict alias.
136     - `@config.json` ? on-disk config; mirrors `AppConfig`. ?
**indexing.default_segmenter='gemini'*** diverges from model default `yolo_hybrid` ? file
wins at runtime.
137     - `@scripts/doctor.py` (formerly root `setup.py`) ? `::init_default_config` ? delegates
to `backend.config.save_default_config` (single-source defaults; skips if config.json exists).
`::run_pip_install` GPU-aware (`nvidia-smi` -> cu124 else cpu). `::run_diagnostics` (Python?3.8,
ffmpeg/ffprobe on PATH, optional torch+CUDA). Run via `python scripts/doctor.py` or `indexer
--setup`.
138
139     ### 2.1 Segmenters (under pipeline/segmenters) ?
140     - `@backend/pipeline/segmenters/base.py`
141     - `::BasePlaySegmenter` ABC ? `__init__(db, config)`; abstract
`name`,`description`,`process_video`. § `process_video(video_id, video_path, player_pool=None,
max_frames=None)` -> List[dict{id,start_time,end_time,duration,metadata}]`.
142     - `::SegmenterRegistry`, `::segmenter_registry` (singleton). ? **register a new
segmenter**: subclass `BasePlaySegmenter`, call `segmenter_registry.register(MyCls)` at module
bottom, AND import the module in `@backend/pipeline/segmenters/__init__.py` (registration is an
IMPORT side-effect). ? `register()` builds `cls(None, {})` just to read `name` ->
name/description/`__init__` MUST NOT touch `self.db`/`self.config`. ? `list_available()` re-
instantiates + swallows exceptions into "No description available." (a throwing `description` is
masked).
143     - `@backend/pipeline/segmenters/__init__.py` ? imports all 6
(`motion,gemini,yolo_hybrid,tracknet_hybrid,wasb_hybrid,fusion_hybrid`) = the de-facto
registration manifest; re-exports them + `segmenter_registry` via `__all__`.
144     - `@backend/pipeline/segmenters/trajectory_hybrid.py::TrajectoryHybridSegmenter` ?
SHARED ENGINE for the trajectory hybrids (2026-06-11 collapse of the copy-paste twins; the "fix
BOTH" footgun is retired): healthcheck gate -> P2 trajectory->windows (+persist) -> P3 provider
handoff -> P4 DB/link; single `::shot_count_from` (old wasb/tracknet shot_count drift gone).
NOT registered itself. ? Subclass contract: `family` (config section
`indexing.<family>.velocity_threshold(0.02)`, `output/<family>` trajectory dir,
`<family>-hybrid-<model>` metadata tag), `display_label` (log lines), `name`/`description`
properties, `_build_runner(idx_cfg)` -> (DetectorRunner, detection_params extras). **Two fusion
seams (IDENTITY by default ? twins byte-identical, pinned by the equivalence test):
`::enrich_candidate_stats(cw,points,fps,frame_width,video_path,idx_cfg)`** (base = the 4
motion-key stats dict; computed once into `window_stats`, used for persist + the handoff gate)
**and `::select_for_handoff(windows>window_stats,idx_cfg)`** (base = all indices). ? shared
knobs still YOLO-named (`log_yolo_candidates/store_yolo_candidates`); whole-video single pass;
needs `{PROVIDER}_API_KEY` (missing -> []). Equivalence pinned by
`tests/test_trajectory_hybrid_equivalence.py` (detector identity, shot_count, offsets, + the
base-hook-identity tests ? parametrized over both twins).
145     - `@backend/pipeline/segmenters/fusion_hybrid.py::FusionSegmenter` (name
`fusion_hybrid`, **default-OFF** multi-signal fusion P2) ? subclasses the engine;
`family='fusion'`, `display_label='Fusion'`. `::build_runner` selects the substrate via
`indexing.fusion.substrate` ('wasb'|'tracknet') reusing the existing runner+config.
`::enrich_candidate_stats` overrides to ALWAYS add `net_crossings` (Gate A signal, GPU-free via
`rally_gate.gate_windows(..., estimate=self._axis_estimate(points))` ? axis/net cached per
trajectory) and, only when `fusion.cue_flags.person`, the ~5 person features
(`::person_features` ? lazy `PersonDetector.detections_per_frame` over the window's ORIGINAL-
video frame range ? `fusion_features.compute_person_features`). `::select_for_handoff`

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overrides to apply served Gate A (`fusion.min_crossings`) ? Gate B (`fusion.min_players`,
person-only) ? default 0/0 ? all windows kept. ? persisted candidates are ALWAYS the full
superset (offline substrate intact); only the handoff/play_segments are gated. ? to A/B vs a
tuned wasb run, set `fusion.velocity_threshold = wasb.velocity_threshold` (the engine reads
`idx_cfg['fusion'].velocity_threshold`). ? person features eager even for gated windows (P3
cleanup). Tests: `tests/test_fusion_hybrid.py`.
146 - `@backend/pipeline/segmenters/wasb_hybrid.py::WasbHybridSegmenter` (name
`'wasb_hybrid'`, the LIVE shuttle substrate; MIT, ships pretrained badminton weights) ? thin
subclass: `family='wasb'`, wires `WasbRunner`/`WasbConfig`; `detection_params` extras =
`weights_path` only.
147 - `@backend/pipeline/segmenters/tracknet_hybrid.py::TrackNetHybridSegmenter` (name
`'tracknet_hybrid'`) ? thin subclass: `family='tracknet'`, wires
`TrackNetRunner`/`TrackNetConfig`; extras = `weights_path` + `queue_length`.
148 - `@backend/pipeline/segmenters/yolo_hybrid.py::HybridYoloSegmenter` (name
`'yolo_hybrid'`, no YOLO remains; ADR-005). **Stage-1 detection EXTRACTED to
`detectors/person_detector.py` (fusion P1, no behavior change)** ? the segmenter now
ORCHESTRATES only (chunking, pipelined-vs-sequential P2 prefetch where ffmpeg overlaps
sequential GPU, AI-provider handoff, candidate persist/link) and delegates detection to
`self.person` (`PersonDetector`). Imports `_DEFAULT_DETECTOR` from person_detector for the
`detection_params` snapshot. ? `yolo_tracker`/`bytetrack.yaml` is dead/back-compat; `gemini_`
config keys are deprecated aliases still honored; still mypy-quarantined for pre-existing
orchestration debt (the torchvision code that caused it left).
149 - `@backend/pipeline/segmenters/gemini.py::GeminiPlaySegmenter` (name `'gemini'`) ?
pure-AI, no P2. Chunks are RE-ENCODED + downsampled to `indexing.ai_chunk_scale_width` (default
1280; 0 = legacy stream-copy) before upload ? stream-copied 4K chunks exceed the provider's
`MAX_UPLOAD_MB` and every chunk gets refused ("0 segments / success"); re-encode also fixes
keyframe drift at chunk boundaries. Oversize refusals (provider metrics `oversize_skipped`) are
aggregated -> `logger.error` + `reporting.run_signals` (`oversize_clips_skipped`) for the report
rule. Workers from `indexing.ai_max_workers`. ? chunks only if `duration>chunk_duration`;
`max_segment_duration=90.0` hardcoded; `debug_duration` silently truncates; `parse_time` accepts
colon timestamps (warns); does NOT persist candidate segments (only hybrids feed the fine-tuning
table).
150 - `@backend/pipeline/segmenters/motion.py::MotionPlaySegmenter` (name `'motion'`) ?
pixel absdiff baseline. ? **metadata + serve/smash/foul/winner events are RANDOM**
(random.sample/randint/uniform/choice); only segmenter using `db.add_event` and passing
server/receiver; uses `print()`; honors `max_frames`.
151
152 ### 2.2 Detectors & windowing (WSL-quarantined; ABC abstraction complete)
153 - `@backend/pipeline/detectors/base.py` ? DetectorRunner ABC (healthcheck() ->
Tuple[bool,str] never-raises; run_predict(...) -> Optional[str] CSV path). Contract deliberately
matches the pre-existing tuple returns and call sites in the hybrids. Rich module docstring
includes "how to add a new runner", Linux-native path notes (the de-risk goal), and guidance on
threading health into reports. Re-exported from `detectors/__init__.py`. This finishes the WSL
de-risk seam (low-regret 6 / review item 1).
154 - `@backend/pipeline/detectors/tracknet_runner.py`
155 - `::TrackNetConfig` (+`.from_indexing_cfg`; ? key drift `wsl_distro` JSON -> `distro`
field), `::TrackNetRunner` ? runs TrackNet `predict.py` in WSL.
156 - `::to_wsl_mnt_path` (`C:\x`->`/mnt/c/x`; ? POSIX/`~` passthrough, UNC unhandled),
`::TrajectoryPoint`(frame,visible,x,y).
157 - **`::parse_trajectory_csv` @staticmethod** + **`::trajectory_to_action_windows`
@staticmethod** = MODEL-AGNOSTIC brain, re-exported verbatim by WasbRunner. ? `weights_path`
defaults `` -> `run_predict` returns None; predict.py only `.mp4/.avi` + hard-names
`<stem>_predict.csv`; `visible==False` resets prev (occlusion gap breaks track); `fps<=0->30`,
`frame_width<=0->1280` silent; `track_id_count` hardcoded 1; `timeout_sec=0` -> NO timeout
(hangs forever). `::stage_video` "OK" sentinel.
158 - `@backend/pipeline/detectors/wasb_runner.py::WasbRunner` (inherits DetectorRunner)
(+`::WasbConfig`, from `indexing.wasb`; ? `weights_path` HAS a real default
`wasb_badminton_best.pth.tar` ? no required guard) ? stages `wasb_infer.py` (`::INFER_WIN`) +
video into WSL, runs in `wasb` env. `::parse_trajectory_csv`/`::trajectory_to_action_windows` =
`staticmethod(TrackNetRunner.*)` (explicit rebind = the plugin-equivalence point). Output hard-
named `<stem>_wasb.csv`. ? `wasb_infer.py` re-staged each run (editing windows copy inert until
staged); `wsl_bash` now shared via `detectors/base.py::WslCommandMixin` (2026-06-11 dedupe).
Both runners now satisfy the DetectorRunner ABC for future swappability. **Cache hygiene
(#109):** on a SUCCESSFUL run both runners delete their WSL intermediates ? WASB the frame cache
+ staged video, TrackNet the staged video ? gated by `indexing.{wasb,tracknet}.keep_frames`

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(default false; on FAILURE kept so a re-run resumes). Pre-run `min_free_gb` disk-guard warning.
Shared helpers `WslCommandMixin::wsl_rm_rf`/`wsl_free_gb`.
159 - `@backend/pipeline/detectors/wasb_infer.py` (WSL) ? `::run` core. ? imports WASB-SBDT
repo modules + torch ? NOT importable on Windows. Reboot-resumable detector cache:
`::DET_FILE='det_raw.jsonl'`, `::MANIFEST_FILE='manifest.json'`, `::CACHE_VERSION=1`,
`::load_and_clean_cache` (reads longest CONTIGUOUS prefix where `w==line index`, rewrites to
heal crash-truncated tail), `::manifest_compatible` (? `frames_dir` abspath'd but `weights`
compared raw), `::atomic_write_text`/`atomic_write_trajectory`, `::dets_to_tracker` (xy MUST be
np.array), `::build_cfg` (`runner.gpus=[0]` hardcoded), `::extract_frames` (cv2-PNG SLOW),
`::default_cache_dir` (`<stem>_wasbcache` next to out). ? GPU pass resumable; CPU tracker
STATEFUL -> always fully re-run; `--fresh` ignores cache.
160 - `@backend/pipeline/detectors/person_detector.py::PersonDetector` ? the **PERSON rally
cue** (torchvision detector + supervision ByteTrack + velocity windowing), extracted from
yolo_hybrid (fusion P1; type-clean, NOT quarantined). `::PERSON_CLASS_ID=1` ? (ultralytics used
0; wrong value -> zero detections),
`::DETECTOR_FACTORIES`/`::DEFAULT_DETECTOR='fasterrcnn_resnet50_fpn'`. `::lazy_load_model`
(lazy `torchvision` import so module/tests load without it; `self.model` typed `Any`),
`::make_tracker` (per-chunk ByteTrack ? IDs chunk-local), `::detect_and_track` (one frame ->
confident persons w/ track IDs; BGR->RGB, COCO person filter, ByteTrack association),
`::extract_action_windows_from_chunk` (chunk -> high-motion windows w/ `peak/mean_velocity`,
`active_frame_count`, `track_id_count` ? the stats that feed candidate `stats` AND the future
learned-fusion person features). ? velocity at `effective_fps = fps/frame_skip` ->
`yolo_velocity_threshold` coupled to `frame_skip`. Consumed by yolo_hybrid AND the fusion
segmenter (P2 reuses the SAME producer over shuttle-seeded windows via `::detections_per_frame`
? add-a-producer, not a refactor). BSD torchvision + MIT ByteTrack.
**`::detections_per_frame(video_path,start_frame,end_frame,frame_skip)`** (fusion P2) reads the
ORIGINAL video by ABSOLUTE frame range (seek+sequential) ? `{frame_idx: [(track_id,bbox)]}` +
`fps/frame_width/frame_height`, so indices align 1:1 with the WASB trajectory `frame` (no clip
re-timing). ? cv2 frame-seek accuracy varies by codec ( $\pm$  a few frames on real footage).
161 - `@backend/pipeline/detectors/fusion_features.py` ? **pure** per-window person-cue
feature math (no I/O/torch/cv2; mirrors rally_gate discipline). Inputs: per-frame
`{frame_idx: [(tid,bbox)]}` + the in-window shuttle points + optional normalized `court_polygon`
+ `net=(axis,pos)`. `::point_in_polygon` (ray-cast), `::person_in_court` (foot-point; None
polygon=count-all; bad dims=False), `::in_court_counts`, `::players_in_court` (median),
`::in_court_ratio`, `::mean_player_motion` (per-track centre displacement; x by width, y by
height), `::two_sided_motion` (both net-halves occupied), `::shuttle_player_colocation` (shuttle
inside a person box  $\pm$  `pad_frac` halo over aligned visible frames ? **the held-shuttle
discriminator**), `::compute_person_features` (aggregator ? the 5 `::PERSON_FEATURE_NAMES`). All
scale/translation-invariant, graceful to None/empty. Tests: `tests/test_fusion_features.py`.
162 - `@backend/pipeline/detectors/rally_gate.py` ? Gate A net-crossing post-filter (pure,
no I/O/model). `::estimate_play_axis` (larger-spread axis; ? ties -> 'x'),
`::count_net_crossings` (sign changes vs median net; deadband jitter zone),
`::gate_windows`/`::apply_gate` (`min_crossings=2` default; kills single-toss service feeds;
**estimate=`param` lets a caller scoring many windows pass a precomputed `(axis,net,span)` to
skip per-window re-estimation ? used by FusionSegmenter).
`::GatedWindow` (start,end,crossings,visible_points,kept). ? window contract here is
(start,end) tuples NOT the rich window dict ? caller must adapt. Consumed by calibrate_local /
distill_local / export_wasb_segments / fusion_hybrid.
163
164 ### 2.3 Eval & calibration (all pure core; I/O in CLI drivers)
165 - `@backend/eval/metrics.py` ? scoring primitive (THE spine). `::interval_iou`,
`::match_intervals` (greedy, deterministic), `::score`->`Score`(`.as_dict()` = wire shape),
`::pr_curve`, `::Match`, `::MatchResult`. ? greedy not Hungarian; empty matches -> 0.0 (not NaN)
for mean_iou/boundary errors.
166 - `@backend/eval/annotations.py` ? generic GT loader. `::load_annotations` (`start,end`
CSV; RAISES on bad rows/`start>=end`), `::parse_timestamp` (decimal or MM:SS/HH:MM:SS),
`::write_scaffold`.
167 - `@backend/eval/calibration.py` ? overfit guard. `::cross_validate` (within-video
k-fold; ? defaults `metric='recall'` deliberately), `::leave_one_video_out` (honest cross-video;
needs 22 labelled videos), `::improvement_report` (OPTIMISTIC ? selected on full GT),
`::select_best`, `::kfold_indices`, `::pct_improvement` (? from-zero -> `inf`), `::evaluate`. $
`PredictFn = Callable[[Config], List[Interval]]` = the plug-in seam. ? `generalization_gap
>~0.1` = overfit alarm.
168 - `@backend/eval/calibrate_wasb.py` ? WASB tuning CLI. `::BASELINE=(0.02,2.0,2.0)`,
`::default_grid` (45 cfgs), `::make_predict_fn` (parses trajectory once, closes over

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`trajectory_to_action_windows`), `::load_golden_gts` ($`start_time,end_time` cols; ? SILENTLY
skips bad rows ? opposite of annotations.load_annotations). ? `{load_golden_gts,
make_predict_fn, BASELINE}` imported by 4 downstream drivers (calibrate_local,
export_wasb_segments, gemini_refine, audio/e1_probe).
169 - `@backend/eval/calibrate_local.py` ? distill Gemini->local thresholds (windowing +
net-crossing gate, no cloud at runtime). `::local_grid` (180 cfigs), `::make_local_predict_fn`
(adds `rally_gate.apply_gate` when `min_crossings>0`), `::build_videos` (? `SystemExit` if a
labels file yields no rallies), `::run`/`::main`. $
`LocalConfig=(velocity,merge,min_dur,min_crossings)`, `BASELINE_LOCAL=(0.02,2.0,2.0,0)`;
manifest JSON shared with distill_local.
170 - `@backend/eval/export_wasb_segments.py` ? tuned cfg -> golden/slicer CSV + comparison
+ optional clip cut. `::TUNED=(0.05,1.0,1.0)` (? from ONE video),
`::SEG_FIELDS`, `::COMP_FIELDS`, `::build_comparison`, `::seconds_to_mmss`, `::maybe_slice`
(lazy-imports rally_slicer). `--slice` requires `--video`. ? `write_segments_csv` output loads
straight into `rally_slicer.load_rally_labels` AND re-imported by gemini_refine ? schema is
load-bearing.
171 - `@backend/eval/batch.py` ? corpus aggregation (pure; orchestration is in
run_phase3_eval). `::aggregate` (micro pool TP/FP/FN, TP-weighted mean_iou, macro F1),
`::default_stages_for` (`auto`; `::FINAL_ONLY_SEGMENTERS={motion,gemini}`),
`::resolve_video_path` (normalizes collector `./data`.. Windows paths), `::format_aggregate`.
172 - `@backend/eval/harness.py` ? DB-pred scorer (no re-run).
`::STAGES=('candidates','final')`, `::get_predictions`
(candidates->`query_candidate_segments(detector=)`; final->`query_play_segments({})`); ? unknown
stage raises ValueError; reused by regression.py),
`::evaluate`->`::EvalReport`(`::StageReport`)->`::report_to_dict` (input contract for
batch.aggregate), `::format_report_text`.
173 - `@backend/eval/training.py` ? learned-segmenter data layer. `::YOLO_FEATURE_NAMES`
(5-d), `::extract_yolo_features` (? reads `row['stats']` dict from DB; missing fields default
0.0), `::label_candidates` (same IoU `match_intervals` as evaluator),
`::build_training_set`->(X,y,intervals); `feature_fn` is the substrate plug-in seam.
174 - `@backend/eval/classifier.py::LogisticRegression` ? numpy-only logreg (no sklearn).
`fit/predict_proba/predict/to_dict/from_dict/save/load`. ? `class_weight='balanced'` default;
stores feature standardization (mean/std) ? inference MUST use same feature ORDER; `_sigmoid`
clips z to [-30,30]; `lr=0.1,epochs=1000,l2=1e-3`; JSON "model":"logreg" tag.
175 - `@backend/eval/eval_partial_labels.py` ? score pipeline vs **PARTIAL human labels**
(PR #60). `::restrict_to_window` (drop preds outside `[0, last GT end]` so un-labeled-tail
detections aren't FPs ? THE partial-label methodology), `::run` (baseline/+gate/tuned/+gate ops
table + per-rally diff via `export_wasb_segments.build_comparison`), `::sweep`/`--sweep` (grid
vs labels; ? fit-on-self = optimistic), `::DEFAULT_GRID`, `::TUNED=(0.05,1.0,1.0)`. ?
`window_end` inferred from `max(GT end)` is WRONG if a video is labelled PAST its last rally ?
P1 fix = collector persists `labeled_window` in the manifest. Reuses calibrate_wasb + rally_gate
+ metrics.
176 - `@backend/features/rally_seq.py` ? **THE single source of truth for the Gen-0
rally_seq per-frame feature schema** (ADR-008): training (`training/`) AND the serving/eval path
import the SAME `build_frame_arrays`/`features` so train==serve by construction. Pure
numpy/scipy (no torch/cv2). `::FEATURE_SCHEMA_VERSION` (=1; bump on ANY semantic change ?
artifacts pin it, loader hard-fails on mismatch), `::FEAT_NAMES` (the 6:
`vis_rate,spd_mean,spd_max,cross_rate,x_std,y_std`), `::FEATURE_SCHEMA`
(`win_sec`/`stride_sec`/`fps_semantics`/`occlusion_rule`),
`::build_frame_arrays(points,frame_width,fps)` (per-frame vis/x/y + per-visible-frame speed &
net-crossings; speed normalized per-SECOND `*fps`, NOT per-frame ? the 2026-06-10 fps-skew fix;
skips speed/crossings across occlusion gaps > `max(2,round(0.25*fps))`),
`::features(arr,fps,win_sec,stride_sec)` (scipy `uniform/maximum_filter1d` windowed -> `X[n,6],
times`)). ? re-exported by `@backend/eval/rally_seq_proto.py`.
177 - `@backend/eval/rally_seq_proto.py` ? **PROTOTYPE** learned per-FRAME rally segmenter =
owned-model **Gen-0 seed** (PR #61). 6 features over the WASB trajectory (`vis_rate, spd_mean,
spd_max, cross_rate, x_std, y_std`) -> `classifier.LogisticRegression`, honest LOVO. ?
`build_frame_arrays`/`features` (+`FEAT_NAMES`) are NOT defined here ? they are **re-exported**
(`noqa F401`) from `@backend/features/rally_seq.py` (ADR-008 single-sourcing; speed per-SECOND +
occlusion-gap-guarded since the 2026-06-10 sweep). Locally-owned: `::labels_for`,
`::probs_to_segments` (smooth->threshold->merge/min-dur), `::roc_auc` (rank-based with **midrank
ties**, scipy.stats.rankdata), `::run` (per-video AUC + LOVO-threshold F1 + oracle-threshold
F1). ? DIRECTIONAL only (n=6, linear, threshold-transfer unsolved); held-out per-frame AUC
0.83?0.92, **beats the heuristic on multi-court footage** (Kushagra/gBaaddy). **n=6 LOVO:
learned 0.626 vs heuristic 0.480** (learned rose from 0.583 after the fps-normalization fix).

```

```

Manifest `output/human_lovo_manifest.json` (n=6; trajectory paths now under `Annotation
Setup/Collect/Trajectories/`) shared w/ calibrate_local.
178 - ? **MULTI-COURT / background-game confound (2026-06-08 finding):** WASB tracks EVERY
shuttle in frame incl. other courts; videos with background games (Kushagra, gBaaddy) have high
dead-time shuttle visibility (41?57% vs 18?22% clean) ? the velocity-windowing heuristic can't
find clean rally boundaries (F1 0.16/0.28 vs ~0.75 on single-court). Diagnostic = in-rally÷dead-
time visibility "contrast" (?3.6× heuristic works, ?1.9× fails). Lever = **foreground-court
isolation** (spatial gate to the prominent court, needs real court-region detection). Academy =
multi-court = the target env. See `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
179 - `@backend/eval/distill_local.py` ? distill Gemini -> a LEARNED classifier (beyond
calibrate_local's threshold ceiling). `::FEATURE_NAMES` (5 fps/res-INVARIANT:
duration,peak,mean,active_ratio,net_crossings), `::build_candidates` (recall-first proposal),
`::predict_rallies` (proba filter then `gemini_refine.merge_overlaps(gap_tol=1.0)`),
`::run`/`::main`. consts `PROPOSAL=(0.005,1.0,0.5)`, `BASELINE_LOCAL=(0.02,2.0,2.0)`. ? leave-
one-video-out needs ?2 videos; final model trained on all.
180 - `@backend/eval/gemini_refine.py` ? refine WASB candidates with Gemini over SHORT
padded clips (precision lever; eval/tuning, NOT the live pipeline).
`::merge_overlaps(gap_tol=0.0)` (reused by distill_local with gap_tol=1.0), `::refine_window` (?
FRESH genai client per worker ? shared throws socket errors), `::full_video_rallies`,
`::run`/`::main` (`--full-video` skips `--trajectory`). ? requires `GEMINI_API_KEY` (raises
SystemExit); imports `BASELINE` from calibrate_wasb +
`TUNED`/`write_segments_csv`/`seconds_to_mmss` from export_wasb_segments.
181 - `@backend/eval/regression.py` ? CI gate. `::DEFAULT_TOLERANCE=0.05`,
`::DEFAULT_BASELINE_PATH=eval_baselines/regression_baseline.json` (tracked, NOT under gitignored
output/), `::regress`, `::regress_with_provider` (GT via ?`annotation_registry.get(tier)`;
"file"->FileProvider/"vendor"->HumanVendor/"auto"->oracle),
`::save_baseline`/`::load_baselines`, `::gt_hash` (label-set fingerprint, reused by
experiment.py), `::RegressionResult`. ? imports `annotation_registry` from
**`backend.annotations`** NOT `backend.eval.annotations`; gate fires on precision OR recall drop
(F1 reported, not gated); no baseline -> `regressed=False` (silent pass). CLI
`run_regression.py` exit 0 ok / 1 regress / 2 no-DB / 3 no-baseline = the promotion gate.
182 - `@backend/eval/experiment.py` ? A/B + shadow + **SxS** variant layer (the thin
"experiment" glue per `docs/EXPERIMENT_HARNESS.md`; composes
metrics/calibration/classifier/training/regression ? NO new scoring math).
`::Variant`(segmenter,config_overrides,cue_flags,`min_crossings`[Gate A],`min_players`[Gate
B],classifier_model,feature_names,threshold,merge_gap;
`to_dict`/`from_dict`/`spec_hash`/`is_learned`), `::CONTROL` (pinned zero-gate baseline).
`::VideoCandidates`(name,intervals,features{col-major},gts), `::predict` (gate?learned-
filter?`gemini_refine.merge_overlaps`), `::evaluate_variant` (static / pinned-model / **honest-
LOVO** train-on-others ? mirrors distill_local), `::compare` (labeled **B?A** deltas +
`label_set_hash`), `::compare_unlabeled` (**NEW-footage SxS**: agreed/a_only/b_only via
`match_intervals`, no GT ? the owner's "SxS on new videos"), `::record_experiment` (?
`eval_baselines/experiment_leaderboard.json`), `::load_video_candidates` (DB bridge via
`query_candidate_segments`). ? enabling a gate needs its feature persisted (absent?0.0?fails
gate, warns); learned variant on an unlabeled video needs a pinned `classifier_model`. **No-reg
invariant (tested): all cues OFF ? byte-identical to CONTROL.**
183
184 - `@backend/eval/golden_fixtures.py` ? **video-free regression harness**
(`docs/GOLDEN_REGRESSION_FIXTURES.md`). `::replay_fixture` =
`tracknet_runner.parse_trajectory_csv` ? `distill_local.build_candidates`(PINNED preset) ?
`experiment.VideoCandidates`(5 shuttle cols only) ? `experiment.predict(CONTROL)` ?
`metrics.score`; **CONTROL / pure-stdlib ? no numpy on the replay path** (cross-OS stable; the
learned/numpy path is deliberately excluded). `::freeze_synthetic` (3 committed synthetic Tier-0
fixtures), `freeze_real_fixture` (now DEFINED in the split-out
`@backend/eval/golden_fixtures.py` (#233) and re-exported here via PEP-562 `__getattr__`)
(P1/M2: `RefusalError` unless `owned`+`minors_free`+non-blank `license`; opaque random
`fx`<hex> slug, metric-invariant time-origin reset, STRICT `gt_loader`, positive
no-`FORBIDDEN_KEYS`/MD5/source-path scan), `::refresh_snapshot`, `::compare_expected` (exact-int
/ approx-1e-6, NEVER byte-compare), `::load_manifest`/`::load_fixture` (schema_version
dispatch). CLI `--freeze-synthetic`/`--freeze-real`/`--refresh-snapshot`/`--check`. Tests:
`tests/test_golden_fixtures.py` (M1/M3 characterization) + `tests/test_golden_real_fixtures.py`
(P1/M2). CI: the `golden-crossos` job (ubuntu/windows/macOS) runs `--check` + the M1 test. ?
synthetic-tier numbers = plumbing correctness, NOT field quality.
185
186 ### 2.3a Audio (E1 probe; NOT adopted for fusion ? ADR-007)

```

```

187     - `@backend/pipeline/audio/hit_detector.py` ? classical-onset shuttle-hit detector. Pure
numpy + stdlib `wave` (NO scipy); ffmpeg only for extraction. `::HitEvent`(t,strength),
`::extract_audio` (ffmpeg -> mono 16k PCM WAV), `::load_wav`, `::onset_envelope` (pure-numpy
STFT spectral flux; `band=(lo,hi)` band-limit), `::detect_hits` (adaptive `mean+k.std`;
`k`=sensitivity knob), `::detect_hits_in_video`. ? separate signal ? WASB never modified by
this; per audio-e1-findings recall 100% but discrimination ~2.2x, band-pass null -> NOT adopted
(PR #35).
188     - `@backend/eval/audio/e1_probe.py` ? E1 GO/NO-GO diagnostic (no training/fusion).
`::cluster_hits` (groups thwacks; ? `min_hits>=2` so a lone service-feed hit isn't a rally),
`::run` (discrimination ratio + recall recovery + audio-only P/R/F1), `::main`. reuses
`hit_detector` + `calibrate_wasb.{load_golden_gts,make_predict_fn,BASELINE}` + `metrics`.
189
190     ### 2.4 Stitcher / tools / utils
191     - `@backend/pipeline/stitcher/compiler.py::VideoCompiler` ? `compile_highlights(raw,
segments:[(start,end)], out_name)`. Per-clip re-encode -> `-f concat -c copy`. `::parse_time`
(mirrors gemini.parse_time; ? unparseable -> 0.0), `::get_video_duration`. ? if duration probe
fails (0.0) NO boundary validation; `begin_padding`/`end_padding` ctor defaults 0.5/1.0 but
every live call site (`/api/compile`, CLI trim, `scripts/run_process_and_stitch.py`) threads
`config.json` stitching.{begin_padding,end_padding} (typed default 0.5/0.5); temp clips in a
TemporaryDirectory under output_dir; raises ValueError/FileNotFoundError/RuntimeError; uses
`print()`. ? two-stage codec contract: clips re-encoded with IDENTICAL settings precisely so
final concat can stream-copy ? change a preset and concat may break. ? optional **branding
watermark** (`stitching.watermark`, default-on): `::watermark_filter` builds a `-vf` (drawtext
via `textfile=` + optional logo `movie/overlay`) applied to the per-clip re-encode only (concat
stays `-c copy`); a failed watermark encode retries the clip WITHOUT it (never nukes the reel,
and logs the ffmpeg stderr reason). Pass `watermark=` to `VideoCompiler(...)` (a
`WatermarkConfig` model or dict; `None` = off). Default font = bundled Apache-2.0
`::BUNDLED_FONT` (`backend/assets/fonts/RobotoSlab-Regular.ttf`) so it renders without
fontconfig; `font_path` overrides.
192     - `@backend/tools/rally slicer.py` ? standalone CLI. `::compute_clip_windows` (pure,
unit-tested), `::load_rally_labels` (golden schema; ? silently skips degenerate rows),
`::slice_rallies`, `::ClipWindow` (rally:str), `::BEGIN_PADDING=0.5`/`::END_PADDING=1.0` (?
duplicate compiler's by copy). `_read_time` -> backend/eval/annotations.parse_timestamp`.
193     - `@tools/generate_highlights.py` ? **the batch highlight RUNNER** (the clean,
orchestrated way to do "make reels for these videos into this folder").
`::generate_highlights_batch(video_paths, out_dir, *, segmenter, local_work_dir, db_path,
force)` + `::atomic_publish` (copy? os.replace; never deletes siblings) + `::resolve_videos`.
**Guarantees:** generates LOCALLY then publishes atomically into the dest (Drive-sync-safe ?
never generate inside a Google-Drive folder); **NEVER bulk-deletes** the dest; per-video
`clear_video_data` (idempotent ? no dup-rally accumulation; pairs with the compiler overlap-
merge); resumable (skip published reels unless `--force`); per-video `` - run.log` +
`_batch_run.log` published additively. CLI: `python -m tools.generate_highlights --videos-dir
<dir> --out <dir> --segmenter gemini [--force]` or repeated `--video`. ? WSL detectors
(wasb/tracknet) localize the source first (Drive invisible to /mnt). **Use THIS for batch reels
? never hand-roll a loop that writes into Drive or rm's the folder** (that lost finished reels +
run.logs once).
194     - `@tools/cleanup_caches.py` ? **top-level** `tools/` ops tool (NOT backend runtime).
Dry-run-first cache GC. Pure `::classify`(name,is_dir,location)/`::plan_deletions` (unit-tested
= the destructive decision path) + I/O `::scan_wsl` (du+find; ? arg-globbing NOT a `for` loop ?
the login shell empties loop vars) / `::scan_windows` / `::delete`. CLI: `python -m
tools.cleanup_caches` (report) . `--apply` / `--keep-video <id|stem>` / `--older-than N` /
`--include-wasbcache` / `--summary` (hook nudge). Purges `*_frames`/staged/proxy/handoff/tmp;
keeps CSV/DB/reels. Companion to the runner self-clean (2.2). See `[[cache-hygiene-policy]]`
memory.
195     - `@backend/utils/video.py` ? `::get_video_duration` (ffprobe; ? 0.0 on failure =
"unknown"), `::extract_video_chunk(...,reencode=False,scale_width=None)`. ? copy mode cuts at
nearest keyframe (drift) ? pass `reencode=True` for short/accurate clips; `scale_width` requires
`reencode=True`.
196     - `@backend/utils/geometry.py` ? `::get_velocity_normalised` (fraction of frame-width/s;
? raw-px fallback if width<=0; used by yolo_hybrid), `::calculate_iou` (? +1 px convention
inflates IoU), `::get_center/get_distance/get_velocity`.
197
198     ### 2.5 Validators ?
199     - `@backend/pipeline/validators/base.py::VideoValidator` ABC, `::ValidationResult`
(pydantic: validator_name,passed,message,details), `::ValidationRegistry` (ordered LIST ?

```

```

registration order = exec order; ? no dedup), `::registry` (singleton). $ `validate(video_path,
config, context) -> ValidationResult`. `::run_all_with_context -> (results, context)` (wraps
each validate in try/except -> failed result not abort); `::run_all` discards context. ? **add a
validator**: subclass + `registry.register(MyValidator())`.
200     - `@backend/pipeline/validators/__init__.py` ? registers in EXEC order: FormatValidator,
ResolutionFPSValidator, PTSContinuityValidator, SceneCutDensityValidator, BlurValidator,
RelevanceValidator (? producers before consumers ? format_check first; ? import mutates global
registry as side effect).
201     - `@backend/pipeline/validators/format.py::FormatValidator` (`format_check`) ?
**PRODUCER** of `context['ffprobe_meta']`; ? MUST run before resolution_fps; container check is
substring `mp4` (so MOV-family passes); external `ffprobe`.
202     - `@backend/pipeline/validators/resolution_fps.py::ResolutionFPSValidator`
(`resolution_fps_check`) ? CONSUMES `ffprobe_meta` (? hard-fails "Run format_check first" if
absent). `min_fps` 29.9; min res 1280x720.
203     - `@backend/pipeline/validators/pts_continuity.py::PTSContinuityValidator`
(`pts_continuity_check`) ? re-probes (no context). ? 10.0s keyframe-gap threshold hardcoded
(comment says 8.0s ? drift, NOT config-driven); any backward jump fails; `<2` keyframes PASSES
(too short).
204     - `@backend/pipeline/validators/scene_cut.py::SceneCutDensityValidator`
(`scene_cut_check`) ? OpenCV. ? `cut_threshold=0.6` hardcoded (NOT config); samples ~2fps, caps
at 100 SAMPLED frames (~first 50s); `max_scene_cuts_per_minute` (0.5) is the config fail
threshold.
205     - `@backend/pipeline/validators/blur.py::BlurValidator` (`blur_check`) ? Laplacian var <
`min_blur_threshold` (default 100.0 in-file, 20.0 in models/config) = fail; samples 15 frames. ?
magic absolute, not resolution-normalized.

```

5. assistant (2026-06-29T10:50:12.900Z)

I have the first file fully. Let me read the rest of CODE_MAP.md to capture the navigation map completely, especially sections beyond line 205 which may cover database, reporting, A/B context, and the "served" operating point.

6. user (2026-06-29T10:50:14.996Z)

```

206     - `@backend/pipeline/validators/relevance.py::RelevanceValidator` (`relevance_check`) ?
HSV court-green ratio (? purely color, no geometry despite name); lazy `import backend.sports`;
3-tier config precedence (`validation.relevance` -> sport profile -> hardcoded); ? unknown sport
-> empty cfg, falls back to defaults, does NOT fail.
207
208     ### 2.6 Sports / Providers / Annotations ?
209     - `@backend/sports/base.py::SportProfile` ABC
(`name`, `get_prompt`, `get_relevance_config`);
`@backend/sports/badminton.py::BadmintonSportProfile` (the Gemini prompt + HSV relevance cfg);
`@backend/sports/__init__.py::sport_registry` (+`::SportRegistry`, auto-registers badminton; ?
`register` instantiates profile to read `name`). ? **add a sport**: subclass `SportProfile`,
`sport_registry.register(MyProfile)`. $ prompt emits JSON array `{start_time(float DECIMAL SEC),
end_time, shuttle_exchange_count(int), shot_count CONFIDENCE('Low'|'Medium'|'High')}`. ? prompt
actively defends against MM:SS; `get()` returns a fresh instance each call.
210     - `@backend/providers/base.py::AIVideoProvider` ABC (`__init__(api_key, model_name)`);
`@backend/providers/gemini.py::GeminiProvider` (lazy `::client` property; `::MAX_UPLOAD_MB=500`;
upload->poll->generate JSON temp 0->delete); `@backend/providers/__init__.py::provider_registry`
(+`::ProviderRegistry`, auto-registers `gemini`). ? **add a provider**: subclass,
`provider_registry.register('name', MyCls)` (key passed EXPLICITLY); `get(name, api_key,
model_name)`. $ `analyze_video(path, prompt, chunk_duration=0.0, label='') -> (segments:list,
metrics:dict{upload_time,process_time,gen_time})`. ? blocking 10s poll; returns `([], metrics)`
on ANY failure (empty != error); oversize clip refused LOUDLY (`logger.error` +
`metrics['oversize_skipped']=True` ? callers aggregate into the report's `oversize_clip_skipped`
rule); orphaned remote file possible if process dies mid-run.
211     - `@backend/annotations/base.py::AnnotationProvider` ABC (`submit/poll/fetch/annotate`),
`::Interval` (= metrics.Interval), `::QUEUED/RUNNING/DONE/FAILED`/`::STATUSES`,
`::AnnotationProviderRegistry` (decorator-friendly `register` returns the class; ? keys off
`provider_cls().name`, falls back to `__name__` if no-arg ctor throws; `list_available()` ->
LIST not dict), `::annotation_registry`. ? **add a tier**: subclass (define `name`),
`annotation_registry.register(MyCls)` at module bottom + import in `__init__.py`. $ `fetch ->
List[Interval]` sorted; contract says RAISE (not `[]`) on unavailable.

```

```

212     - `@backend/annotations/file_provider.py::FileProvider` (`file`; job_id IS the
resolved CSV path; `_resolve` -> `

```

```

222 - `@backend/reporting/spec.yaml` ? $ declarative report layout + footgun engine consumed
by obsreport. `::customer_verdict` (pass/pass_with_warnings/fail messages). `::sections`
(ordered `{id,title,kind,audience?{both,customer,technical}}`). ? `::rules` ? each
`{code,severity,when[{path,op,value}],message,faq_ref,customer_message?}`; `when` clauses AND'd;
`path` dotted-navigates the recorder JSON. ? `path` strings are a HARD coupling to harvest's
stage/metric/detail names ? renaming a detail in harvest.py silently breaks the matching rule.
Rules (9): `silent_provider_noop` (error: no api key + 0 segs), `oversize_clip_skipped` (error:
provider refused chunks over MAX_UPLOAD_MB ? 4K stream-copy footgun),
`ai_empty_vs_no_rallies` (warn), `fps_fallback_used` (warn), `audio_disabled_or_missing` (warn: no
audio stream -> loses the shuttle-thwack recall cue; reads `video_meta.audio_present`),
`synthetic_motion_metadata` (warn), `segmenter_divergence` (info),
`reingest_replaced_prior` (info), `validation_failed` (error: halts indexing).
223 - `@backend/reporting/run_signals.py` ? ? in-process, thread-safe channel pipeline-
internals -> report (`::incr(video_id,key,by)`, `::pop(video_id)`). No obsreport dep (always
importable). Segmenter worker threads record facts the harvest can't see (e.g.
`oversize_clips_skipped`); `emit_indexer_report` pops UNCONDITIONALLY (even reporting-off, so
nothing accumulates) and passes to `record_run`. ? process-local best-effort: signals are
dropped (by design) if no report is emitted for that video_id.
224
225 ---
226
227 ## 3. DATA CONTRACTS (canonical shapes ? one place)
228
229 $ **Trajectory CSV** (produced by tracknet/wasb predict, consumed by
`parse_trajectory_csv`):
230 `Frame,Visibility,X,Y` ? header+column order fixed; `Visibility==1`=visible; X,Y pixel
coords.
231 -> `TrajectoryPoint{frame:int, visible:bool, x:float, y:float}` (frame-sorted).
232
233 $ **Candidate window dict** (produced by `trajectory_to_action_windows` AND yolo
`extract_action_windows_from_chunk`):
234 `{start:float(sec), end:float(sec), active_frame_count:int, peak_velocity:float,
mean_velocity:float, track_id_count:int(==1 for trajectory), chunk_index:Optional[int]}`.
235
236 $ **candidate_segments DB row** (`db.add_candidate_segment` /
`query_candidate_segments`):
237 `{id, video_id, detector:str, chunk_index, start_time:REAL, end_time:REAL,
duration:REAL, confidence:REAL(min(1.0,peak_velocity)),
stats:JSON{peak_velocity,mean_velocity,active_frame_count,track_id_count},
detection_params:JSON{...}, confirmed_segment_id, human_label, notes, created_at}`
(stats/detection_params deserialized on read).
238 `detection_params` keys by detector:
wasb=`{detector,velocity_thresh,merge_gap,min_action_window_duration,weights_path}`; tracknet
adds `queue_length`; yolo=`{velocity_thresh,merge_gap,frame_skip,tracker,detector_model,detector
_score_threshold,min_action_window_duration}`.
239
240 $ **play_segment dict** (`db.add_play_segment` / `query_play_segments`):
241 `{id:int, video_id, start_time:float, end_time:float, duration:float(end-start), server,
receiver, metadata:dict, events:[...]}`. Hybrid P4 metadata `{shot_count,
shot_count_confidence:'Unknown', detector:f'{family}-hybrid-{model}'}`; gemini detector = bare
`model_name`; motion adds random `intensity/avg_speed_kmh`.
242
243 $ **jobs row** (`db.create_job` / `get_job` ? #16, PR-1): `{id:TEXT(uuid4.hex),
video_path, video_id?:TEXT(set once hashed), segmenter?, player_pool?:JSON list, max_frames?,
status:'queued'|'running'|'done'|'failed' (EXECUTION state ? pipeline verdict is
result["status"]), progress?:TEXT(stage label), error?:TEXT, result?:JSON(the former sync
/api/process response incl. reports), created_at, started_at?, finished_at?}`.
244
245 $ **AI segment dict** (provider output, consumed by hybrid P3 / gemini): `{start_time,
end_time, shuttle_exchange_count?, shot_count_confidence?}` (offset by chunk/window start;
`_candidate_id` stamped internally).
246
247 $ **GT / golden CSVs** (TWO conventions ? do not conflate):
248 - generic (`annotations.load_annotations`): `start,end` 2-col; optional header; decimal
or MM:SS/HH:MM:SS; RAISES on bad row.

```



```

249     - golden/slicer (`calibrate_wasb.load_golden_gts`, `rally_slicer.load_rally_labels`,
`export.SEG_FIELDS`): `rally_number,start_time,end_time[,duration,start_mmss,end_mmss]`;
SILENTLY skips bad rows.
250
251     $ **det_raw.jsonl** (one window/line): `{ "w":<window_index>,
"f":[[frame_id,[[x,y,score,scale],...]], ...] }`. Lines MUST be contiguous (line N has `w==N`);
first violation/JSONDecodeError -> truncate-heal.
252     $ **manifest.json** (cache): `{version, frames_dir(abspath), weights, sport, frames_in,
frames_total, windows_total, windows_done}`. Reuse only if ALL match + `version==CACHE_VERSION`.
253     $ **collector manifest.json** (eval): `{eval_sets:[{video_id(label), local_path, csv,
...}]}`; ? DB key = file MD5, manifest `video_id` is a label.
254
255     $ **eval primitives**: `Interval=Tuple[float,float]` (start<end).
`Match{pred_index,gt_index,iou}`. `MatchResult{matches, unmatched_pred_indices(=FP),
unmatched_gt_indices(=FN)}`. `Score{iou_threshold,num_pred,num_gt,true_positives,false_positives
,false_negatives,precision,recall,f1,mean_iou,mean_start_error,mean_end_error}.as_dict()`.
256
257     $ **WSL detector candidate** (in-WSL, model<->tracker): `{ 'xy':np.array([x,y]),
'score':float, 'scale':int }`; plain JSON form `[x,y,score,scale]`. ? `_dets_to_tracker` MUST
rebuild `xy` as np.array (tracker does `np.linalg.norm`).
258
259     $ **HitEvent** (audio, `pipeline/audio/hit_detector`): `{t:float(sec), strength:float}`
? onset-envelope peak.
260
261     $ **observability report** (`reporting/harvest.record_run` -> `RunRecorder`): stages
`validation/segmentation/ai_handoff` + highlights + verdict ? {pass,pass_with_warnings,fail};
flags `{code,faq_ref}`; `video_meta.fps_source` ? {probed,fallback}. ? stage/metric/detail names
are coupled to `spec.yaml::rules[].when[].path` ? pairs that MUST stay in sync:
`ai_handoff.details.{api_key_present,ai_based}`, `segmentation.metrics.segment_count`, `segmenta
tion.details.{segmenter_actual,matches_config_default,segmenter_explicitly_requested,was_reinges
t}`, `validation.status`, `video_meta.fps_source`.
262
263     ---
264
265     ## 4. WASB WSL CONTRACT (external dep `~/models/WASB-SBDT`, MIT; NOT in this repo ?
claims below are unverified against external source, only the repo-side caller wasb_infer.py was
checked)
266     - Env: WSL conda `wasb`, cwd `~/models/WASB-SBDT/src`. Hydra
`compose(config_name='eval', overrides=[dataset=<sport>, model=wasb,
detector.model_path=<weights>, runner.device=cuda, runner.gpus=[0]])`. `gpus=[0]` because box is
single-GPU. (overrides VERIFIED in @backend/pipeline/detectors/wasb_infer.py::build_cfg.)
267     - `model=wasb` -> `name=hrnet`, `frames_in=3`, `frames_out=3`, input 512x288 (WxH),
ImageNet normalize; affine resize maps heatmap coords -> original-frame coords. (model dims read
from cfg["model"] at runtime in wasb_infer.run.)
268     - `detectors/detector.py::TracknetV2Detector.run_tensor(imgs, affine_mats) -> (results,
hms_vis)`. `results[bid][eid]` = list of `{xy:np.array (ORIGINAL coords), score, scale}`.
(caller uses `detector.run_tensor(imgs, trans)` returning `(batch_results, _)` ? consistent.)
269     - `trackers/online.py::OnlineTracker` ? STATEFUL, sequential per `update()` (assumes
every frame fed once, in order, no gaps). `update(frame_dets)->{x,y,visi,score}`; `refresh()`
resets. `det['xy']` MUST be np.array. Full ordered re-run is deterministic. (caller calls
`build_tracker(cfg)`, `tracker.refresh()`, `tracker.update(...)` returning dict with `visi/x/y`
? consistent.)
270     - `dataloaders/dataset_loader.py::ImageDataset(cfg, samples, input_wh, output_wh,
transform, seq_transform, is_train)`; `samples=[{frames:[paths], annos:[{frame_path,
center:center(is_visible,x,y)}]}]`. (caller constructs exactly this ? VERIFIED in
wasb_infer.run.)
271     - Weights `pretrained_weights/wasb_badminton_best.pth.tar` ? NOT committed, academic-
data-trained -> ? confirm terms before COMMERCIAL deploy. monotrack weights = noncommercial,
avoided.
272
273     ---
274
275     ## 5. CROSS-REPO (sibling private repo `sports-data-collector` ? external, unverified
here)
276     - Indexer keys video by **FILE MD5**; collector keys by human `video_id`. ? re-

```

```

encoding/re-downloading changes bytes -> MD5; acquisition MUST precede processing; manifest
tracks the exact file used.
277     - `python -m src.cli.curate export` (collector) -> `<out>/<sport>/<video_id>.csv`
(indexer `start,end` format) + `manifest.json` -> consumed by `@backend/eval/batch.py` +
`@scripts/run_phase3_eval.py`.
278     - Collector also emits golden rally labels (`rally_number,start_time,end_time,...`) ->
consumed by `@backend/eval/calibrate_wasb.py` / `export_wasb_segments.py` /
`@backend/tools/rally_slicer.py`.
279
280     ---
281
282     ## 6. GOTCHAS (cross-cutting footguns)
283     - **MD5 keying:** `video_id` = MD5 of WHOLE file via the single
`@backend/utils/hashing.py::compute_video_id` helper (64KB chunks), imported by main + all
run_*.py. Manifest `video_id` is a label, NOT the DB key.
284     - **Config access:** `backend/main.py` and all four `run_*.py` load via
`backend.config.load_config` (typed `AppConfig`)). The legacy `.get()` dict-compat shim remains
for validators/segmenters. Remaining literal: a defensive `getattr(..., 'motion')` fallback in
`main.py` that only fires if `global_config`/`.indexing` is `None` (the ASGI-import edge case).
285     - **Segmenter default:** single source of truth = model
`IndexingConfig.default_segmenter='yolo_hybrid'; `config.json` may override at runtime (ships
`gemini`). run_*.py read the model value (no literals); `main.py` keeps the defensive `'motion'`
getattr fallback noted above.
286     - **`AppConfig.get` must return dict sections, not models:** the legacy
`.get("indexing",{ }).get(...)` chains everywhere depend on it; the `extra="allow"` model also
silently keeps typo'd config keys (no rejection).
287     - **`output_dir`:** a real field on `OutputConfig` (default `"output"`); the CLI
`--output-dir` overrides it on the typed model in `main()`, and `compile_highlights` reads the
same `global_config.output.output_dir` ? honored end-to-end.
288     - **Gemini free-tier / API-key dead path:** hybrids + gemini return `[]` (not error) if
`{PROVIDER}_API_KEY` missing OR WSL healthcheck fails ? a silent no-op. `analyze_video` also
returns `[]` on real failure -> empty != "no rallies". The reporting
`silent_provider_noop`/`ai_empty_vs_no_rallies` rules exist precisely to disambiguate this in
the report.
289     - **Reporting never breaks a run:** `emit_indexer_report` swallows all exceptions
(designed for `main.py` `finally:`) and is emitted even on validation failure. `obsreport` is a
SOFT dep ? absent -> no-op.
290     - **spec.yaml <-> harvest.py coupling:** rule `path` strings dotted-navigate harvest's
stage/metric/detail names; rename a detail in harvest.py and the matching rule silently stops
firing. Add rules in spec.yaml, not code.
291     - **fps / frame_width must be PROBED not assumed:** `_probe_fps_width` / runner
fallbacks (30.0, 1280.0) silently mis-scale velocity thresholds; calibrate_wasb CLI bakes
`59.94/1920`. The report's `fps_fallback_used` flag surfaces this.
292     - **WSL `/mnt` can't see Google Drive / network mounts:** stage video into native WSL fs
(`~/clips`) ? `/mnt/c` reads are slow and Drive-backed paths invisible.
293     - **ffmpeg NOT in WSL** by design: all ffmpeg/ffprobe run Windows-side (must be on
PATH); WSL only runs the GPU model. `extract_frames` uses cv2 PNG (slow) ? prefer host ffmpeg +
`--frames_dir`. ? Don't import `wasb_infer` from Windows code (it imports torch + WASB repo
modules).
294     - **GPU float nondeterminism:** detector pass is checkpointed/resumable but not bit-
reproducible; the CPU tracker re-run over the ordered cache IS deterministic.
295     - **`output/` is gitignored:** DB, baselines, cache, reels live there. Predictions MUST
exist in `output/<db_name>` before any eval/regression.
296     - **Validator ordering coupling:** exec order is Format -> ResolutionFPS ->
PTSContinuity -> SceneCut -> Blur -> Relevance; `resolution_fps_check` hard-fails unless
`format_check` ran earlier (shared `context['ffprobe_meta']`). PTS 10.0s gap + scene-cut 0.6
correlation are NOT config-overridable.
297     - **`add_video` is an UPSERT** (`INSERT ... ON CONFLICT(id) DO UPDATE`) that mutates the
row in place ? re-ingesting a `video_id` does NOT cascade-delete its segments (the old `INSERT
OR REPLACE` + FK-cascade wipe is gone). Segment clearing is now an explicit destroy-AFTER-
confirm via `segment_watermarks`/`prune_segments` (or `clear_video_data`). `_get_connection`
MUST still re-issue `PRAGMA foreign_keys=ON` per connection (CASCADE/SET NULL still apply on an
explicit delete).
298     - **Twin segmenters:** `wasb_hybrid.py` / `tracknet_hybrid.py` are copy-paste; fixes to
P3/P4 must land in both. ? subtle drift: wasb's shot_count `int(None)`-via-TypeError fallback vs

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tracknet's `is None` check.
299     - **Two "annotations" modules:** `backend.eval.annotations` (GT CSV loader) vs
`backend.annotations` (provider registry). regression.py imports the latter.
300     - **`calibrate_wasb.{load_golden_gts, make_predict_fn, BASELINE}` is a de-facto shared
API** for calibrate_local, export_wasb_segments, gemini_refine, e1_probe ? changing its golden-
CSV schema or `BASELINE` ripples widely. `gemini_refine.merge_overlaps` reused by
`distill_local`; `harness.get_predictions` reused by `regression.regress`.
301     - **Hardcoded tuned constants:** `calibrate_wasb.BASELINE`, `export.TUNED`,
slicer/compiler padding ? duplicated by copy, not import; re-tuning won't propagate.
302     - **`timeout_sec=0` = no timeout:** a hung WSL/GPU call blocks forever.
303     - **Stitch concat assumes identical encode:** per-clip re-encode then `-c copy` concat;
if probe fails (0.0) no boundary validation.
304     - **Audio NOT in the live pipeline:** `pipeline/audio/hit_detector` +
`eval/audio/e1_probe` are a diagnostic probe (ADR-007); recall 100% but discrimination ~2.2x ->
not adopted for fusion.
305
306     ---
307
308     ## 7. RAMP PATHS (to do X -> open Y::symbol)
309     | Task | Open |
310     |---|---|
311     | Change a config default / add a config field | `@backend/config/models.py` (single
source of truth) -> regenerate via `scripts/doctor.py`/`save_default_config`; ? config.json may
override (default_segmenter) |
312     | Read config in new code | `from backend.config import load_config, AppConfig`
(`@backend/config/__init__.py`); attribute access, NOT raw `json.load` |
313     | Tune candidate windowing (velocity/merge/min-dur) |
`@backend/pipeline/detectors/tracknet_runner.py::trajectory_to_action_windows`; cfg `config.json
indexing.{wasb,tracknet}.velocity_threshold`, `dead_time_merge_gap`,
`min_action_window_duration` |
314     | Add a segmenter | subclass `@backend/pipeline/segmenters/base.py::BasePlaySegmenter`;
`segmenter_registry.register(...)` at module bottom; import in
`@backend/pipeline/segmenters/__init__.py`; if AI-based add to
`@backend/reporting/harvest.py::AI_SEGMENTERS` |
315     | Change stitching padding | `config.json stitching.{begin_padding,end_padding}`; impl
`@backend/pipeline/stitcher/compiler.py::VideoCompiler.compile_highlights`; wiring
`@backend/main.py` (`api/compile` + CLI trim); mirror
`@backend/tools/rally_slicer.py::BEGIN_PADDING/END_PADDING` |
316     | Reclaim disk / clean caches | `python -m tools.cleanup_caches` (dry-run) then
`--apply`; runs also self-clean via `indexing.{wasb,tracknet}.keep_frames=false` (default) +
`min_free_gb` guard; tool `@tools/cleanup_caches.py` |
317     | Filter highlights by shot count | `config.json stitching.min_shot_count` ? enforced on
BOTH `@backend/main.py::compile_highlights` (`api/compile`) and
`@scripts/run_process_and_stitch.py` (segments without `shot_count` metadata are exempt) |
318     | Add a sport | subclass `@backend/sports/base.py::SportProfile`;
`sport_registry.register(...)` in `@backend/sports/__init__.py`; define prompt+HSV in new file |
319     | Add an AI provider | subclass `@backend/providers/base.py::AIVideoProvider`;
`provider_registry.register('name', Cls)` |
320     | Add an annotation/GT tier | subclass
`@backend/annotations/base.py::AnnotationProvider`; register + import in
`@backend/annotations/__init__.py` |
321     | Add / change a report footgun rule | `@backend/reporting/spec.yaml::rules`
(declarative; ? `when.path` must match harvest's stage/metric/detail names); facts in
`@backend/reporting/harvest.py` |
322     | Tweak what the report records | `@backend/reporting/harvest.py::record_run` + stage
builders; emit point is `@backend/main.py` `finally:` ->
`@backend/reporting/__init__.py::emit_indexer_report` |
323     | Run calibration (WASB thresholds) | `@backend/eval/calibrate_wasb.py::main`
(`--trajectory --labels --fps --frame-width`); trust `cv` recall not `fit` |
324     | Distill Gemini -> local (thresholds / learned) |
`@backend/eval/calibrate_local.py::main` (thresholds) OR `@backend/eval/distill_local.py::main`
(learned classifier); both `--manifest` |
325     | Refine WASB candidates with Gemini (tuning) | `@backend/eval/gemini_refine.py::main`
(`--full-video` skips `--trajectory`; needs `GEMINI_API_KEY`) |
326     | Export tuned rally segments / clips | `@backend/eval/export_wasb_segments.py::main`

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(`--slice` needs `--video`) |
327 | Export timelines (EDL/JSON/CSV) + basic stats (Value Layer slice) |
`backend/tools/export.py` (segments_to_edl/json/csv, basic_stats); called from anywhere that has
play_segments (CLI, notebook, future API) |
328 | Run a video end-to-end | `@backend/main.py::main` (`--video-path` or `--server`) or
`@scripts/run_process_and_stitch.py::main` (edit hardcoded paths) |
329 | Score DB preds vs GT (one video) | `@run_eval.py::main` |
330 | Batch score a collector manifest | `@scripts/run_phase3_eval.py::main` (`--manifest`)
|
331 | Guard a regression (CI) | `@run_regression.py::main` (`--video-id`); exit 1=regress,
2=missing DB; snapshot via `--update-baseline` |
332 | Guard rally-quality nightly (no GPU) | `@scripts/nightly_regression.py::main` ? re-
score golden corpus (default + milestone configs) vs `eval_baselines/nightly_baseline.json`;
exit 1=regress, 4=stale; `--update-baseline` to re-snapshot after an intended change. Scheduled
via `scripts/register_nightly_task.ps1` |
333 | Guard the FULL pipeline + reel-count nightly (GPU) |
`@scripts/nightly_reelcount.py::main` ? run the real configured+checkpointed pipeline on gs://
clips, assert reel-count EXACT + quality + report cost, email it; exit 1=regress, 4=stale;
`--update-baseline` to re-freeze. Runs on an ephemeral GCP GPU VM
(`deploy/gcp/nightly_regression/launch.sh`) that self-deletes |
334 | Run the whole test suite | `@run_all_tests.py` (repo root); or `pytest tests/`
directly |
335 | Resume a crashed WASB pass | nothing to do ?
`@backend/pipeline/detectors/wasb_infer.py::run` auto-resumes from
`det_raw.jsonl`+`manifest.json` next to `--out`; `--fresh` forces restart |
336 | Add a validator | subclass `@backend/pipeline/validators/base.py::VideoValidator`;
`registry.register(instance)` in `@backend/pipeline/validators/__init__.py` (mind order;
producers before consumers) |
337 | Add a DB column/table | `@backend/infrastructure/database.py::Database._init_db`
migration ladder + bump `PRAGMA user_version` (else version-assert tests fail) |
338 | Train the learned segmenter | `@backend/eval/training.py::build_training_set` ->
`@backend/eval/classifier.py::LogisticRegression.fit/save` |
339 | Probe an audio signal (diagnostic) | `@backend/eval/audio/e1_probe.py::main`; detector
in `@backend/pipeline/audio/hit_detector.py` |
340
341 ---
342
343 ## 8. TEST MAP (`tests/test_*.py` -> subsystem)
344 All pure-logic / mocked-boundary unit tests; no WSL/GPU/Gemini/ffmpeg actually invoked.
`@run_all_tests.py` (repo root) is the suite runner.
345 | Test | Covers |
346 |---|---|
347 | `test_base.py` | `pipeline/segmenters/base.SegmenterRegistry` (register/get/list via
DummySegmenter) |
348 | `test_yolo_hybrid.py` | `pipeline/segmenters/yolo_hybrid.HybridYoloSegmenter`
ORCHESTRATION (mocks the `self.person` cue seams; candidate link; `store_yolo_candidates=False`
skip; output-unchanged = the fusion-P1 regression guard) |
349 | `test_person_detector.py` | `pipeline/detectors/person_detector.PersonDetector`
(`PERSON_CLASS_ID` filter + ByteTrack handoff; `extract_action_windows_from_chunk` enriched
dicts; `detections_per_frame` per-frame map + height) |
350 | `test_fusion_features.py` | `pipeline/detectors/fusion_features` (pure feature math:
point-in-polygon, in-court counts, players/ratio, motion, two-sided w/ polygon, colocation halo
+ alignment) |
351 | `test_fusion_hybrid.py` | `pipeline/segmenters/fusion_hybrid.FusionSegmenter`
(net_crossings persisted; superset; Gate A & Gate B handoff gating; person-cue features;
`_build_runner` substrate select) |
352 | `test_tracknet_hybrid.py` |
`pipeline/segmenters/tracknet_hybrid.TrackNetHybridSegmenter` (4-phase; healthcheck-abort;
`_probe_fps_width` fallback) |
353 | `test_gemini.py` | `pipeline/segmenters/gemini.GeminiPlaySegmenter` (single vs chunked
path, per-chunk offset, `shot_count`<-`shuttle_exchange_count`, chunk re-
encode/`ai_chunk_scale_width` args, oversize-skip -> run_signals + ERROR log) |
354 | `test_tracknet_runner.py` | `pipeline/detectors/tracknet_runner` (`to_wsl_mnt_path`,
`from_indexing_cfg`, parse + `trajectory_to_action_windows` shared brain, healthcheck,
run_predict, stage) |

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355 | `test_wasb_runner.py` | `pipeline/detectors/wasb_runner.WasbRunner` WSL-glue
(DetectorRunner conformance, healthcheck, `parse_trajectory_csv`/`trajectory_to_action_windows`
rebind, run_predict; WSL mocked) |
356 | `test_rally_gate.py` | `pipeline/detectors/rally_gate` (axis/net estimate, crossing
count, gate kills single-toss) |
357 | `test_wasb_cache.py` | `pipeline/detectors/wasb_infer` resumable cache (serialize,
contiguity heal, manifest invalidation, atomic writes, `default_cache_dir`) |
358 | `test_eval_metrics.py` | `eval.metrics` IoU/matching/score/pr_curve |
359 | `test_eval_annotations.py` | `eval.annotations` GT CSV parse/validate,
`write_scaffold` |
360 | `test_eval_harness.py` | `eval.harness` DB-pred fetch + evaluate + report_to_dict;
also `@run_eval.py::main`/`get_file_md5` (exit codes, `--scaffold`/`--json`) |
361 | `test_eval_batch.py` | `eval.batch` path resolve / stage select / aggregate
(micro/macro, zero-div) |
362 | `test_calibration.py` | `eval.calibration`
pct_improvement/kfold/select_best/cross_validate/LOVO/improvement_report |
363 | `test_calibrate_wasb.py` | `eval.calibrate_wasb`
load_golden_gts/make_predict_fn/grid/run, BASELINE |
364 | `test_calibrate_local.py` | `eval.calibrate_local` local grid + crossing gate, leave-
one-video-out |
365 | `test_export_wasb_segments.py` | `eval.export_wasb_segments` segment/comparison CSV
schemas, mmss; export-> slicer reuse |
366 | `test_eval_training.py` | `eval.training` label_candidates / feature extraction /
build_training_set |
367 | `test_eval_classifier.py` | `eval.classifier.LogisticRegression`
fit/predict_proba/balanced/JSON round-trip |
368 | `test_eval_regression.py` | `eval.regression` baseline store/compare/gate; also
`@run_regression.py::main` (exit-code CI gate) |
369 | `test_distill_local.py` | `eval.distill_local` build_candidates / predict_rallies /
learned-vs-baseline |
370 | `test_gemini_refine.py` | `eval.gemini_refine` merge_overlaps / _offset_segments (pure
helpers, no API) |
371 | `test_hit_detector.py` | `pipeline/audio/hit_detector` (detect_hits, onset_envelope,
_moving_stats) + `eval/audio/e1_probe.cluster_hits` |
372 | `test_compiler.py` | `pipeline/stitcher/compiler.VideoCompiler` (`_parse_time`
formats, re-encode->concat path, boundary validation; ffmpeg/ffprobe mocked) |
373 | `test_rally_slicer.py` | `tools.rally_slicer.compute_clip_windows` (pure) + label
parse |
374 | `test_video.py` | `utils.video` duration/chunk (subprocess mocked) |
375 | `test_geometry.py` | `utils.geometry` center/distance/velocity/normalised-velocity/IoU
|
376 | `test_database.py` | `infrastructure.database.Database`
schema/CRUD/migrations/user_version/EXPECTED_CANDIDATE_COLUMNS guard |
377 | `test_config.py` | `config` (load_config defaults, save_default_config, AppConfig
defaults pinned, `extra` tolerance, `.get()` dict-compat) |
378 | `test_reporting_harvest.py` | `reporting` (`emit_indexer_report`,
`harvest.record_run`); ? `importorskip("obsreport")`; all 9 footgun rules + plain-dict-OR-
AppConfig + `report.enabled=False` |
379 | `test_run_signals.py` | `reporting/run_signals` (incr/pop consume + per-video
isolation; emit consumes signals even with reporting OFF). ? deliberately NO obsreport dep ?
this is the half of the oversize channel CI can verify (obsreport is absent there) |
380 | `test_reporting_spec_faq.py` | `reporting/spec.yaml` <-> `README.md` cross-ref
integrity (every `faq_ref` resolves; actionable rules cite a `faq_ref`); ?
`importorskip("obsreport")` |
381 | `test_validators_base.py` | `pipeline/validators/base.ValidationRegistry`
(register/run_all/run_all_with_context -> `(results,context)`; exception captured not raised)
|
382 | `test_sport_profiles.py` | `sports` registry + BadmintonSportProfile (prompt contains
"decimal seconds"/chunk value; relevance config keys) |
383 | `test_providers.py` / `test_provider_gemini.py` | `providers` registry +
GeminiProvider lifecycle (`MAX_UPLOAD_MB` guard refuses LOUDLY w/ `oversize_skipped` metric,
non-list JSON -> [], cleanup delete) |
384 | `test_annotation_providers.py` / `test_automated_provider.py` | `annotations`
registry, FileProvider, AutomatedProvider (oracle/lineage); keystone one-scorer contract ->
`eval.metrics.score` |

```

385 | (no dedicated test) | individual `pipeline/validators/\*` impls, `main.py`,
`scripts/run\_process\_and\_stitch.py`, `pipeline/segmenters/{wasb\_hybrid,motion}`,
`eval/audio/e1\_probe` (diagnostic) ?? exercised only indirectly |
386

7. assistant (2026-06-29T10:50:21.383Z)

I now have both files fully read. Both exist. I have a complete picture of the rally-detection pipeline, the navigation map, and the A/B "served" operating point concept. Let me compile the structured answer.

The "served" operating point appears in several places:

- `nightly\_regression.py` evaluates TWO configs: `default` (SERVED, milestone knobs OFF) + `milestone` (R3 cap=6 + R4 inactivity trim)
- `experiment.py::CONTROL` (pinned zero-gate baseline) is the A side
- `fusion\_hybrid.py` defaults all gates OFF so it is byte-identical to CONTROL
- The "No-reg invariant (tested): all cues OFF ? byte-identical to CONTROL"

Let me return the structured output.

8. user (2026-06-29T10:51:16.620Z)

Structured output provided successfully